ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jun 2025 P1HR Worked Solutions

Jun 2025 | P1HR

33 questions ▪ question image followed by worked solution

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STUDENT EDITION

Worked answers included

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PAST PAPER QUESTION

Q#: 1 Marks: 2

TOPIC: **Prime Factors, HCF & LCM**

Jun 2025 P1HR - Jun 2025 | P1HR

1 Find the lowest common multiple (LCM) of 45 and 70

(Total for Question 1 is 2 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 2

TOPIC: **Prime Factors, HCF & LCM**

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WORKED SOLUTION

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1. Find the LCM

$$45 = 3^2 \times 5$$

$$70 = 2 \times 5 \times 7$$

2. Find the LCM

For the LCM, take the highest power of each prime:

$$\text{LCM} = 2 \times 3^2 \times 5 \times 7 = 630$$

FINAL ANSWER

630

PAST PAPER QUESTION

Q#: 2 **Marks: 2**

TOPIC: **Rounding, Estimation & Bounds**

Jun 2025 P1HR - Jun 2025 | P1HR

2 The length of a ship is 142.8 m, correct to 1 decimal place.

(i) Write down the lower bound of the length of the ship.

.... m

(1)

(ii) Write down the upper bound of the length of the ship.

.... m

(1)

(Total for Question 2 is 2 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 2

TOPIC: **Rounding, Estimation & Bounds**

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WORKED SOLUTION

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1. The length is m correct

The length is 142.8 m correct to 1 decimal place.

2. Half of is

Half of 0.1 is 0.05.

$$142.8 - 0.05 = 142.75$$

$$142.8 + 0.05 = 142.85$$

FINAL ANSWER

(i) 142.75 m, (ii) 142.85 m

PAST PAPER QUESTION

Q#: 3 Marks: 3

TOPIC: **Algebraic Proof**

Jun 2025 P1HR - Jun 2025 | P1HR

3 Show that $2\frac{1}{4} \times 1\frac{5}{7} = 3\frac{6}{7}$

(Total for Question 3 is 3 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 3

TOPIC: **Algebraic Proof**

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WORKED SOLUTION

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1. Evaluate fraction

$$2\frac{1}{4} = \frac{9}{4}, \qquad 1\frac{5}{7} = \frac{12}{7}$$

$$2\frac{1}{4} \times 1\frac{5}{7} = \frac{9}{4} \times \frac{12}{7}$$

$$= \frac{108}{28} = \frac{27}{7} = 3\frac{6}{7}$$

FINAL ANSWER

$3\frac{6}{7}$.

PAST PAPER QUESTION

Q#: 4 Marks: 5

TOPIC: **Probability Toolkit**

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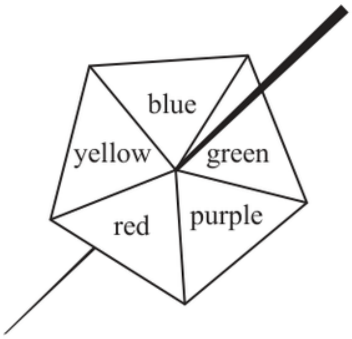
4 Here is a biased 5-sided spinner.
When the spinner is spun, it can land on blue or on green or on purple or on red or on yellow.



The table gives information about the probability of the spinner landing on each colour.

Colour	blue	green	purple	red	yellow
Probability	0.12	0.20	0.38	$4x$	x

Sophie spins the spinner once.

(a) Work out the probability that the spinner lands on blue or on green or on purple.

....
(1)

Max spins the spinner 350 times.

(b) Work out an estimate for the number of times the spinner lands on red.

....
(4)

(Total for Question 4 is 5 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 5

TOPIC: **Probability Toolkit**

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WORKED SOLUTION

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1. Let the yellow probability be

Let the yellow probability be x. Then the red probability is $4x$.

$$0.12 + 0.20 + 0.30 + 4x + x = 1$$

$$5x = 0.38$$

$$x = 0.076$$

2. Calculate probability

So

$$P(\text{red}) = 4x = 0.304$$

$$P(\text{blue, green or purple}) = 0.12 + 0.20 + 0.30 = 0.62$$

3. Expected red

Expected red:

$$350 \times 0.304 = 106.4$$

FINAL ANSWER

0.62, and about 106.

PAST PAPER QUESTION

Q#: 5 Marks: 4

TOPIC: **Set Notation & Venn Diagrams**

Jun 2025 P1HR - Jun 2025 | P1HR

5 $\mathscr{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

$A = \{2, 4, 6, 8, 10, 12\}$

$B = \{3, 6, 9, 12\}$

$C = \{1, 3, 5, 7, 9, 11\}$

(a) List the members of the set

(i) $A \cup B$

....

(ii) B'

....

(2)

$\mathscr{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

$A = \{2, 4, 6, 8, 10, 12\}$

$B = \{3, 6, 9, 12\}$

$C = \{1, 3, 5, 7, 9, 11\}$

$\mathscr{E}$	$\cap$	$\cup$	$\varnothing$	$\in$	$\notin$

(b) Write a symbol from the box on each dotted line to make each of the following a true statement.

(i) $A \cap C =$ ....

(ii) 13 .... $\mathscr{E}$

(2)

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 4

TOPIC: **Set Notation & Venn Diagrams**

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WORKED SOLUTION

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1. Calculate value

$$\mathcal{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

$$A = \{2, 4, 6, 8, 10, 12\}$$

$$B = \{3, 6, 9, 12\}$$

$$C = \{1, 3, 5, 7, 9, 11\}$$

2. Calculate value

Therefore

$$A \cup B = \{2, 3, 4, 6, 8, 9, 10, 12\}$$

3. Calculate value

and

$$B' = \{1, 2, 4, 5, 7, 8, 10, 11\}$$

4. No number is both even

No number is both even and odd, so

$$A \cap C = \emptyset$$

5. Set

Also $13 \notin \mathcal{E}$ because 13 is not in the universal set.

FINAL ANSWER

(a)(i) $\{2, 3, 4, 6, 8, 9, 10, 12\}$, (a)(ii) $\{1, 2, 4, 5, 7, 8, 10, 11\}$, (b)(i) $\emptyset$, (b)(ii) $13 \notin \mathcal{E}$

PAST PAPER QUESTION

Q#: 6 Marks: 4

TOPIC: **Solving Inequalities**

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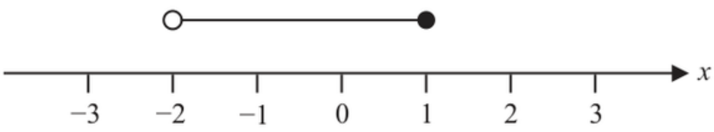
6



(a) Write down the inequality shown on the number line.

....

(2)

(b) Solve the inequality $7a - 5 \leqslant 3a + 28$

Show clear algebraic working.

....

(2)

(Total for Question 6 is 4 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 4

TOPIC: **Solving Inequalities**

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1. (a) The number line has

(a) The number line has an open circle at -2, a closed circle at 1, and shading between them.

$$-2 < x \leq 1$$

2. (b)

(b)

$$7a - 5 \leq 3a + 28$$

$$4a \leq 33$$

$$a \leq \frac{33}{4}$$

FINAL ANSWER

$-2 < x \leq 1$, and $a \leq \frac{33}{4}$.

PAST PAPER QUESTION

Topic: **Standard & Compound Units**

Jun 2025 P1HR - Jun 2025 | P1HR

Q#: 7 Marks: 3

7 Change a speed of $50x$ metres per second to a speed in kilometres per hour.

.... km/h

(Total for Question 7 is 3 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 3

TOPIC: **Standard & Compound Units**

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WORKED SOLUTION

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1. Convert m/s to km/h multiply

To convert m/s to km/h, multiply by 3.6.

$$50x \times 3.6 = 180x$$

FINAL ANSWER

$180x$ km/h.

PAST PAPER QUESTION

Q#: 8 **Marks: 5**

TOPIC: **Factorising**

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8 (a) Simplify $a^6 \times a^{10}$

....

(1)

(b) Simplify $c^{30} \div c^{12}$

....

(1)

(c) (i) Factorise $y^2 - 10y + 21$

....

(2)

(ii) Hence, solve $y^2 - 10y + 21 = 0$

....

(1)

(Total for Question 8 is 5 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 5

TOPIC: **Factorising**

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WORKED SOLUTION

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1. (a)

(a) $a^6 \times a^{10} = a^{16}$.

2. (b)

(b) $c^{30} \div c^{12} = c^{18}$.

3. (c)(i)

(c)(i)

$$y^2 - 10y + 21 = (y-3)(y-7)$$

4. (c)(ii)

(c)(ii)

$$y = 3 \quad \text{or} \quad y = 7$$

FINAL ANSWER

a^{16}, c^{18}, $(y-3)(y-7)$, $y=3$ or $y=7$

PAST PAPER QUESTION

Q#: 9 **Marks: 3**

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

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9 The diagram shows triangle *PQR*

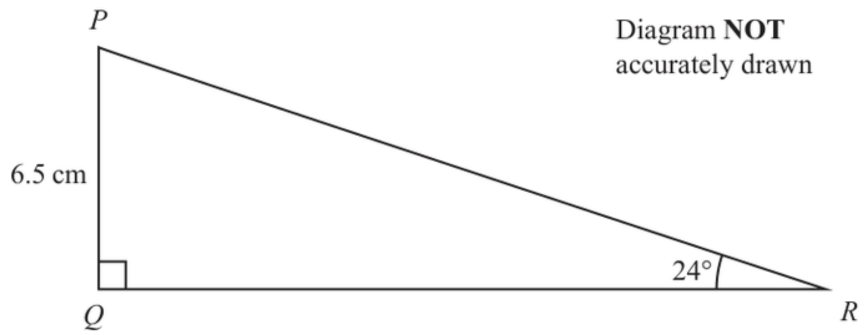

Diagram **NOT** accurately drawn

Work out the length of *QR*
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 9 is 3 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 3

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

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WORKED SOLUTION

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1. Use trigonometry

$$\tan 24^\circ = \frac{6.5}{QR}$$

$$QR = \frac{6.5}{\tan 24^\circ} = 14.599\ldots$$

FINAL ANSWER

14.6 cm.

PAST PAPER QUESTION

Q#: 10 Marks: 3

TOPIC: **Volume & Surface Area**

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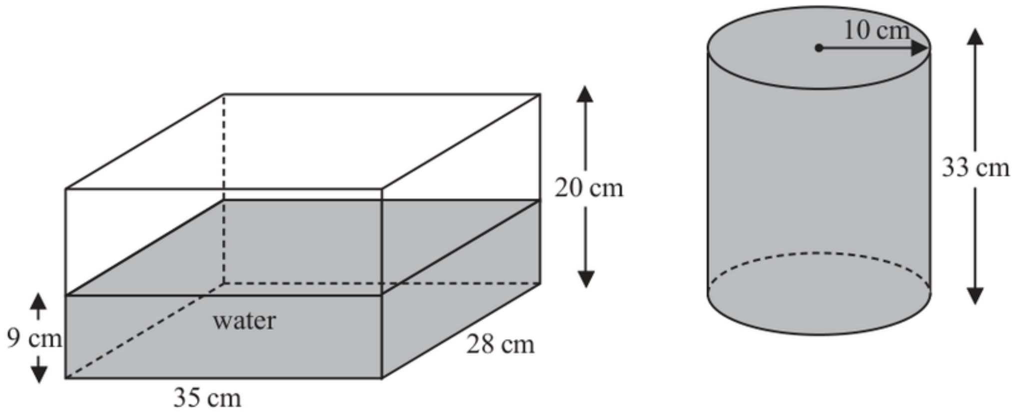
10 The diagram shows two water containers.
One is a cuboid and one is a cylinder.

Diagram **NOT** accurately drawn



The cuboid measures 35 cm by 28 cm by 20 cm
The surface of the water in the cuboid is 9 cm above the base of the cuboid.

The cylinder has a radius of 10 cm and a height of 33 cm
The cylinder is completely full of water.

Izzy is going to pour all the water from the cylinder into the cuboid.

Show that the cuboid will not be completely full of water.

(Total for Question 10 is 3 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 3

TOPIC: **Volume & Surface Area**

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WORKED SOLUTION

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1. Capacity of the cuboid

Capacity of the cuboid:

$$35 \times 28 \times 20 = 19600 \text{ cm}^3$$

2. Water already in the cuboid

Water already in the cuboid:

$$35 \times 28 \times 9 = 8820 \text{ cm}^3$$

3. Volume of the full cylinder

Volume of the full cylinder:

$$\pi(10)^2(33) = 10367.255\ldots \text{ cm}^3$$

4. Total water after pouring

Total water after pouring:

$$8820 + 10367.255\ldots = 19187.255\ldots$$

5. Calculate area

Since

$$19187.255\ldots < 19600$$

6. the cuboid will not be

the cuboid will not be completely full.

FINAL ANSWER

The cuboid will not be completely full.

PAST PAPER QUESTION

Q#: 11 Marks: 5

Topic: **Compound Interest & Depreciation**

Jun 2025 P1HR - Jun 2025 | P1HR

11 Zhou invests some money for 2 years.
He invests \$2500 with Bank A and \$3000 with Bank B.

Bank A

Invests \$2500

amount of money invested : total amount of interest after 2 years = 20 : 3

Bank B

Invests \$3000

4% per year compound interest for 2 years

Zhou receives more **interest** from Bank A than from Bank B.

How much more?

\$ ....

(Total for Question 11 is 5 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 5

TOPIC: **Compound Interest & Depreciation**

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WORKED SOLUTION

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1. Use compound interest

For Bank A,

$$\text{amount invested} : \text{interest} = 20 : 3$$

2. Use compound interest

So the interest from Bank A is

$$2500 \times \frac{3}{20} = 375$$

3. Use compound interest

For Bank B, the compound interest after 2 years is

$$3000(1.04)^2 - 3000$$

$$= 3000(1.0816) - 3000 = 244.80$$

4. Difference

Difference:

$$375 - 244.80 = 130.20$$

FINAL ANSWER

\$130.20

PAST PAPER QUESTION

Q#: 12 Marks: 6

TOPIC: **Cumulative Frequency Diagrams**

Jun 2025 P1HR - Jun 2025 | P1HR

12 The table gives information about the ages of 80 people in a cinema.

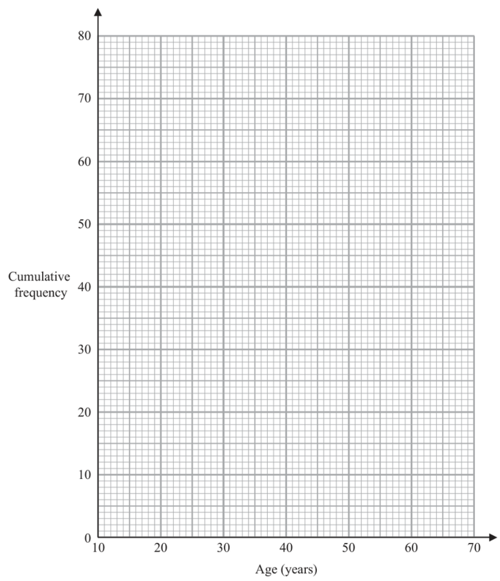
Age (n years)	**Frequency**
$10 < n \leqslant 20$	12
$20 < n \leqslant 30$	15
$30 < n \leqslant 40$	20
$40 < n \leqslant 50$	18
$50 < n \leqslant 60$	9
$60 < n \leqslant 70$	6

(a) Complete the cumulative frequency table.

Age (n years)	**Cumulative frequency**
$10 < n \leqslant 20$	
$10 < n \leqslant 30$	
$10 < n \leqslant 40$	
$10 < n \leqslant 50$	
$10 < n \leqslant 60$	
$10 < n \leqslant 70$	

(1)

(b) On the grid below, draw a cumulative frequency graph for your table.



(2)

(c) Use your graph to find an estimate for the percentage of the 80 people who are more than 46 years of age.
Give your answer correct to the nearest whole number.

.... %

(3)

(Total for Question 12 is 6 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 6

TOPIC: **Cumulative Frequency Diagrams**

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WORKED SOLUTION

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1. The cumulative frequencies are

The cumulative frequencies are

$$12,\ 27,\ 47,\ 65,\ 74,\ 80$$

2. Plot

Plot

$$(10,0),(20,12),(30,27),(40,47),(50,65),(60,74),(70,80)$$

3. Use cumulative frequency

At 46 years,

$$F(46) \approx 47 + \frac{6}{10}(65-47) = 57.8$$

4. Convert standard form

So the number older than 46 is

$$80 - 57.8 = 22.2$$

$$\frac{22.2}{80} \times 100 \approx 27.75\%$$

FINAL ANSWER

about 28%.

PAST PAPER QUESTION Q#: 13 Marks: 2

TOPIC: **Statistics Toolkit**

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13 Here are the numbers of eggs laid by some hens on each of 11 days in June.

9	10	12	15	17	18	19	19	20	25	26

Work out the interquartile range of the numbers of eggs.

(Total for Question 13 is 2 marks)

PAST PAPER SOLUTION Q#: 13 Marks: 2

TOPIC: **Statistics Toolkit**

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WORKED SOLUTION

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1. Put the numbers in order

Put the numbers in order:

$$9,\ 10,\ 12,\ 15,\ 17,\ 18,\ 19,\ 19,\ 20,\ 25,\ 26$$

$$Q_1 = 12, \qquad Q_3 = 20$$

$$\text{IQR} = 20 - 12 = 8$$

FINAL ANSWER

8.

PAST PAPER QUESTION

Q#: 14 Marks: 2

TOPIC: **Powers, Roots & Standard Form**

Jun 2025 P1HR - Jun 2025 | P1HR

14 Work out $6.7 \times 10^{135} + 3 \times 10^{134}$

Give your answer in standard form.

(Total for Question 14 is 2 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 2

TOPIC: **Powers, Roots & Standard Form**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

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1. Convert standard form

$$6.7 \times 10^{135} + 3 \times 10^{134}$$

2. Write both terms using

Write both terms using 10^{135}:

$$3 \times 10^{134} = 0.3 \times 10^{135}$$

3. Convert standard form

So

$$6.7 \times 10^{135} + 0.3 \times 10^{135} = 7.0 \times 10^{135}$$

$$= 7 \times 10^{135}$$

FINAL ANSWER

7×10^{135}

PAST PAPER QUESTION

Q#: 15 **Marks: 6**

TOPIC: **Solving Linear Equations**

Jun 2025 P1HR - Jun 2025 | P1HR

15 (a) Solve $\dfrac{5a+8}{3} - \dfrac{2a+5}{4} = 23$

Show clear algebraic working.

$a =$ ....

(4)

(b) Express $\left(\dfrac{\sqrt{y}}{3}\right)^{-1}$ in the form cy^n where c and n are numbers to be found.

....

(2)

(Total for Question 15 is 6 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 6

TOPIC: **Solving Linear Equations**

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WORKED SOLUTION

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1. Solve equation

For part (a),

$$\frac{5a+8}{3} - \frac{2a+5}{4} = 23$$

2. Multiply by

Multiply by 12:

$$4(5a+8) - 3(2a+5) = 276$$

$$20a + 32 - 6a - 15 = 276$$

$$14a + 17 = 276$$

$$14a = 259$$

$$a = \frac{259}{14} = \frac{37}{2}$$

3. Solve equation

For part (b),

$$\left(\frac{\sqrt{y}}{3}\right)^{-1} = \frac{3}{\sqrt{y}}$$

$$= 3y^{-1/2}$$

FINAL ANSWER

(a) $a = \frac{37}{2}$, (b) $3y^{-1/2}$

PAST PAPER QUESTION

Q#: 16 Marks: 2

TOPIC: **Algebraic Proof**

Jun 2025 P1HR - Jun 2025 | P1HR

16 Use algebra to show that the recurring decimal $0.6\dot{1}\dot{2} = \dfrac{101}{165}$

(Total for Question 16 is 2 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 2

TOPIC: **Algebraic Proof**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

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1. Let

Let

$$x = 0.6121212\ldots$$

2. Calculate value

Then

$$10x = 6.121212\ldots$$

3. Calculate value

and

$$1000x = 612.121212\ldots$$

4. Subtract

Subtract:

$$1000x - 10x = 612.121212\ldots - 6.121212\ldots$$

$$990x = 606$$

$$x = \frac{606}{990} = \frac{101}{165}$$

FINAL ANSWER

$0.6121212\ldots = \frac{101}{165}$.

PAST PAPER QUESTION Q#: 17 Marks: 3

TOPIC: **Algebraic Proof**

Jun 2025 P1HR - Jun 2025 | P1HR

17 Prove that, for any three numbers which are consecutive multiples of 4, the difference between the square of the largest number and the square of the smallest number is always a multiple of 64

Show clear algebraic working.

(Total for Question 17 is 3 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 3

TOPIC: **Algebraic Proof**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

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1. Let the three consecutive multiples

Let the three consecutive multiples of 4 be

$$4n, \quad 4n+4, \quad 4n+8$$

2. The difference between the square

The difference between the square of the largest and the square of the smallest is

$$(4n+8)^2 - (4n)^2$$

$$= 16n^2 + 64n + 64 - 16n^2$$

$$= 64n + 64 = 64(n+1)$$

3. This is always a multiple

This is always a multiple of 64.

FINAL ANSWER
The difference is always a multiple of 64.

PAST PAPER QUESTION

Q#: 18 Marks: 5

TOPIC: **Direct & Inverse Proportion**

Jun 2025 P1HR - Jun 2025 | P1HR

18 F is inversely proportional to the cube of r

$F = 6$ when $r = 2$

(a) Find a formula for F in terms of r

....

(3)

(b) Find the value of r when $F = 3072$

$r =$ ....

(2)

(Total for Question 18 is 5 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 5

TOPIC: **Direct & Inverse Proportion**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

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1. Find inverse function

F is inversely proportional to r^3, so

$$F = \frac{k}{r^3}$$

2. Use when

Use $F = 6$ when $r = 2$:

$$6 = \frac{k}{2^3}$$

$$k = 48$$

3. Find inverse function

So

$$F = \frac{48}{r^3}$$

4. Find inverse function

When $F = 3072$,

$$3072 = \frac{48}{r^3}$$

$$r^3 = \frac{48}{3072} = \frac{1}{64}$$

$$r = \frac{1}{4}$$

FINAL ANSWER

$F = \frac{48}{r^3}$, and $r = \frac{1}{4}$.

PAST PAPER QUESTION

TOPIC: **Probability Toolkit**

Jun 2025 P1HR - Jun 2025 | P1HR

Q#: 19 Marks: 3

19 Kannika has 9 counters.
There is a number on each counter.

1 1 2 3 3 3 5 6 6

Kannika puts the 9 counters in a bag.
She takes at random a counter from the bag and does not replace the counter.
She then takes at random a second counter from the bag.

Work out the probability that the sum of the numbers on the two counters is less than 5

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 3

TOPIC: **Probability Toolkit**

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WORKED SOLUTION

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1. The counters are

The counters are

$$1,\ 1,\ 2,\ 3,\ 3,\ 3,\ 5,\ 6,\ 6$$

2. There are

There are

$$\binom{9}{2} = 36$$

3. ways to choose two counters

ways to choose two counters.

4. Calculate probability

For a sum less than 5, the pairs are:

$$(1,1), \quad (1,2), \quad (1,3)$$

5. Counting the repeated counters gives

Counting the repeated counters gives

$$1 + 2 + 6 = 9$$

6. Calculate probability

So

$$P(\text{sum} < 5) = \frac{9}{36} = \frac{1}{4}$$

FINAL ANSWER

$\frac{1}{4}$.

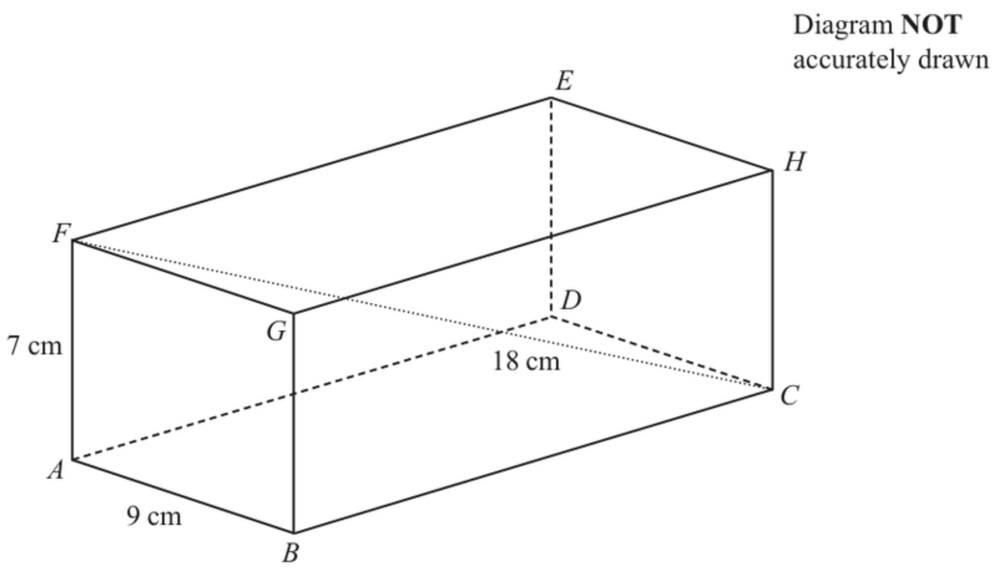
PAST PAPER QUESTION

Q#: 20 Marks: 3

TOPIC: **3D Pythagoras & Trigonometry**

Jun 2025 P1HR - Jun 2025 | P1HR

20 The diagram shows cuboid *ABCDEFGH*



$AB = 9\,\text{cm}$ $\quad AF = 7\,\text{cm}$ $\quad FC = 18\,\text{cm}$

Calculate the length of BC
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 3

TOPIC: **3D Pythagoras & Trigonometry**

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WORKED SOLUTION

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1. Use trigonometry

In the rectangle $ABCF$, use Pythagoras with the diagonal FC:

$$AC^2 = FC^2 - AF^2$$

$$AC^2 = 18^2 - 7^2 = 324 - 49 = 275$$

2. Use trigonometry

Now use Pythagoras in triangle ABC:

$$BC^2 = AC^2 - AB^2$$

$$BC^2 = 275 - 9^2 = 275 - 81 = 194$$

$$BC = \sqrt{194} = 13.928\ldots$$

FINAL ANSWER

13.9 cm to 3 significant figures

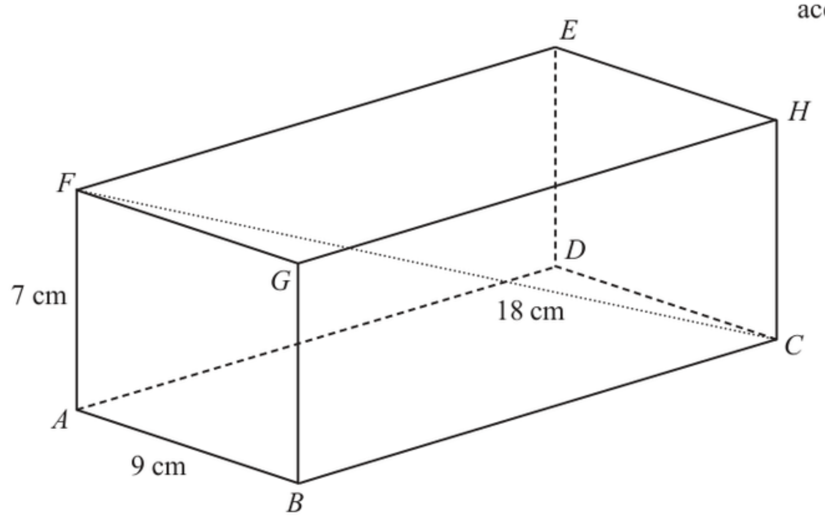
PAST PAPER QUESTION Q#: 20 Marks: 3

TOPIC: **3D Pythagoras & Trigonometry**

Jun 2025 P1HR - Jun 2025 | P1HR

20 The diagram shows cuboid *ABCDEFGH*

Diagram **NOT** accurately drawn



$AB = 9\,\text{cm}$ $AF = 7\,\text{cm}$ $FC = 18\,\text{cm}$

Calculate the length of *BC*
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 3

TOPIC: **3D Pythagoras & Trigonometry**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Use trigonometry

In the rectangle $ABCF$, use Pythagoras with the diagonal FC:

$$AC^2 = FC^2 - AF^2$$

$$AC^2 = 18^2 - 7^2 = 324 - 49 = 275$$

2. Use trigonometry

Now use Pythagoras in triangle ABC:

$$BC^2 = AC^2 - AB^2$$

$$BC^2 = 275 - 9^2 = 275 - 81 = 194$$

$$BC = \sqrt{194} = 13.928\ldots$$

FINAL ANSWER

13.9 cm to 3 significant figures

PAST PAPER QUESTION Q#: 21 Marks: 3

Topic: **Transformations of Graphs**

Jun 2025 P1HR - Jun 2025 | P1HR

21 The curve with equation $y = \mathrm{f}(x)$ has one turning point.
The coordinates of this turning point are $(-6, 9)$

(a) Write down the coordinates of the turning point on the curve with equation $y = \mathrm{f}(3x)$

(.................... , ...)
(1)

The curve **C** with equation $y = \mathrm{g}(x)$ is transformed to give the curve **S** with equation $y = \mathrm{g}(x + a) + b$

The point $(4, -5)$ on **C** is mapped to the point $(1, -16)$ on **S**

(b) Write down the value of a and the value of b

$a =$ ....

$b =$ ....
(2)

(Total for Question 21 is 3 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 3

TOPIC: **Transformations of Graphs**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. The turning point of is

The turning point of $y = f(x)$ is $(-6, 9)$.

2. Calculate value

For

$$y = f(3x)$$

3. the coordinate is divided by

the x-coordinate is divided by 3:

$$(-6, 9) \to (-2, 9)$$

4. Calculate value

For part (b), a point $(4, -5)$ on $y = g(x)$ maps to $(1, -16)$ on

$$y = g(x + a) + b$$

5. Calculate value

For the x-coordinate:

$$1 + a = 4$$

$$a = 3$$

6. Calculate value

For the y-coordinate:

$$-5 + b = -16$$

$$b = -11$$

FINAL ANSWER

$(-2, 9)$, $a = 3,\ b = -11$

PAST PAPER QUESTION

Q#: 21 **Marks: 3**

Topic: **Transformations of Graphs**

Jun 2025 P1HR - Jun 2025 | P1HR

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$a =$ ....

$b =$ ....

(2)

(Total for Question 21 is 3 marks)

PAST PAPER SOLUTION Q#: 21 Marks: 3

TOPIC: **Transformations of Graphs**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

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For the y-coordinate:

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$$b = -11$$

FINAL ANSWER

$(-2, 9)$, $a = 3$, $b = -11$

PAST PAPER QUESTION

Q#: 22 Marks: 5

TOPIC: **Simultaneous Equations**

Jun 2025 P1HR - Jun 2025 | P1HR

22 Solve the simultaneous equations

$$x^2 + y^2 = 41$$
$$2x + y = 3$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION Q#: 22 Marks: 5

TOPIC: **Simultaneous Equations**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Calculate value

$$x^2 + y^2 = 41$$

$$2x + y = 3$$

2. Calculate value

So

$$y = 3 - 2x$$

3. Substitute

Substitute:

$$x^2 + (3 - 2x)^2 = 41$$

$$x^2 + 9 - 12x + 4x^2 = 41$$

$$5x^2 - 12x - 32 = 0$$

$$(5x + 8)(x - 4) = 0$$

4. Evaluate fraction

So

$$x = -\frac{8}{5} \quad \text{or} \quad x = 4$$

5. Find

Find y:

$$x = -\frac{8}{5} \Rightarrow y = \frac{31}{5}$$

$$x = 4 \Rightarrow y = -5$$

FINAL ANSWER

$(x, y) = \left(-\frac{8}{5}, \frac{31}{5}\right)$ or $(4, -5)$

PAST PAPER QUESTION

Q#: 22 Marks: 5

TOPIC: **Simultaneous Equations**

Jun 2025 P1HR - Jun 2025 | P1HR

22 Solve the simultaneous equations

$$x^2 + y^2 = 41$$
$$2x + y = 3$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 5

TOPIC: **Simultaneous Equations**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

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FINAL ANSWER

$(x, y) = \left(-\frac{8}{5}, \frac{31}{5}\right)$ or $(4, -5)$

PAST PAPER QUESTION Q#: 23 Marks: 5

Topic: **Circle Theorems**

Jun 2025 P1HR - Jun 2025 | P1HR

23

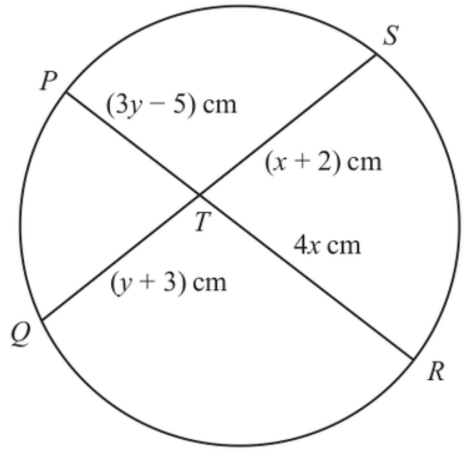


Diagram **NOT** accurately drawn

PTR and *QTS* are chords of a circle.

$PT = (3y - 5)$ cm $\quad QT = (y + 3)$ cm $\quad RT = 4x$ cm $\quad ST = (x + 2)$ cm

Find an expression for y in terms of x

$y = $ ....

Total for Question 23 is 5 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 5

TOPIC: **Circle Theorems**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Use chord theorem

For two intersecting chords in a circle,

$$PT \times TR = QT \times TS$$

2. Substitute the given lengths

Substitute the given lengths:

$$(3y-5)(4x) = (y+3)(x+2)$$

3. Expand

Expand:

$$12xy - 20x = xy + 2y + 3x + 6$$

4. Collect the terms on one

Collect the y terms on one side:

$$12xy - xy - 2y = 20x + 3x + 6$$

$$11xy - 2y = 23x + 6$$

5. Factorise

Factorise:

$$y(11x-2) = 23x + 6$$

$$y = \frac{23x+6}{11x-2}$$

FINAL ANSWER

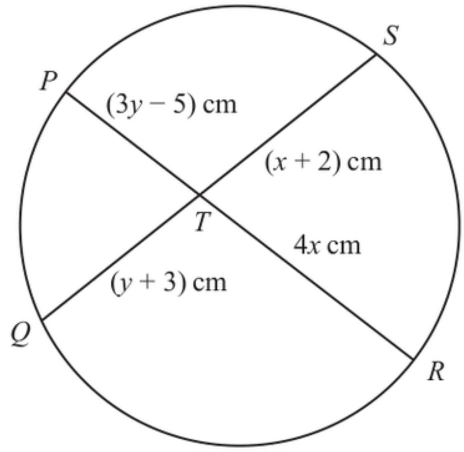
$$\boxed{y = \frac{23x+6}{11x-2}}$$

PAST PAPER QUESTION Q#: 23 Marks: 5

TOPIC: **Circle Theorems**

Jun 2025 P1HR - Jun 2025 | P1HR

23



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Find an expression for y in terms of x

$y =$ ....

Total for Question 23 is 5 marks)

PAST PAPER SOLUTION Q#: 23 Marks: 5

TOPIC: **Circle Theorems**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

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5. Factorise

Factorise:

$$y(11x - 2) = 23x + 6$$

$$y = \frac{23x + 6}{11x - 2}$$

FINAL ANSWER

$$\boxed{y = \frac{23x + 6}{11x - 2}}$$

PAST PAPER QUESTION

Q#: 24 Marks: 6

TOPIC: **Sequences**

Jun 2025 P1HR - Jun 2025 | P1HR

24 The first 3 terms of an arithmetic series are

$$(2x + 5) \qquad (3y - 4) \qquad (4x - 2)$$

where x and y are constants.

The sum of the first 9 terms of the series is 216

Find the value of x and the value of y
Show clear algebraic working.

$x =$ ....

$y =$ ....

(Total for Question 24 is 6 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 6

TOPIC: **Sequences**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. The first 3 terms are

The first 3 terms are

$$2x + 5, \quad 3y - 4, \quad 4x - 2$$

2. Equal differences give

Equal differences give

$$(3y - 4) - (2x + 5) = (4x - 2) - (3y - 4)$$

$$6y - 6x = 11$$

3. Use series formula

For an arithmetic series,

$$S_9 = 9(a + 4d)$$

4. Here

Here

$$a = 2x + 5$$

$$d = (3y - 4) - (2x + 5) = 3y - 2x - 9$$

5. Given

Given $S_9 = 216$:

$$9[(2x + 5) + 4(3y - 2x - 9)] = 216$$

$$12y - 6x = 55$$

6. Solve

Now solve

$$6y - 6x = 11$$

$$12y - 6x = 55$$

7. Subtracting gives

Subtracting gives

$$6y = 44$$

$$y = \frac{22}{3}$$

8. Evaluate fraction

Then

$$6\left(\frac{22}{3}\right) - 6x = 11$$

$$x = \frac{11}{2}$$

FINAL ANSWER

$x = \frac{11}{2}$, $y = \frac{22}{3}$

PAST PAPER QUESTION

Q#: 24 Marks: 6

TOPIC: **Sequences**

Jun 2025 P1HR - Jun 2025 | P1HR

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Show clear algebraic working.

$x =$ ....

$y =$ ....

(Total for Question 24 is 6 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 6

TOPIC: **Sequences**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

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FINAL ANSWER

$x = \frac{11}{2}$, $y = \frac{22}{3}$

PAST PAPER QUESTION

Q#: 25 Marks: 6

TOPIC: **Circles, Arcs & Sectors**

Jun 2025 P1HR - Jun 2025 | P1HR

25

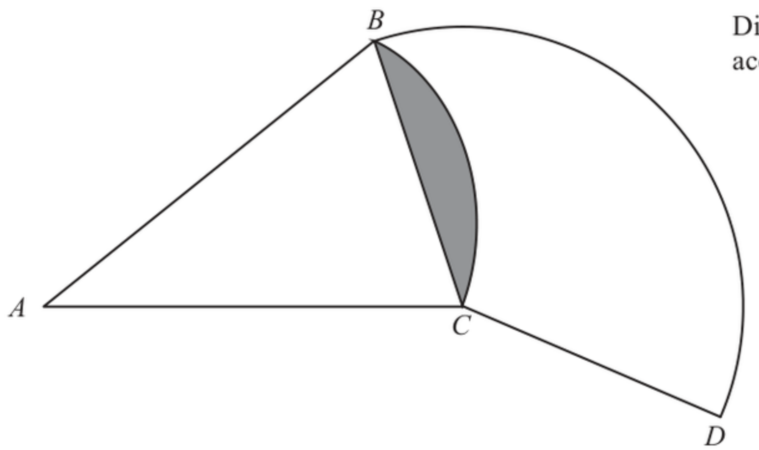

Diagram **NOT** accurately drawn

BAC is a sector of a circle, centre *A*

BCD is a sector of a circle, centre *C*

Angle $BAC = 40°$
Angle $BCD = 130°$
Area of shaded segment $= 28\,\text{cm}^2$

Find the length of the arc *BD*
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 25 is 6 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 6

TOPIC: **Circles, Arcs & Sectors**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Let the radius of sector

Let the radius of sector BAC be R.

2. The shaded segment is the

The shaded segment is the segment cut off by chord BC in the sector with centre A.

3. Its area is

Its area is

$$\text{sector area} - \text{triangle area}$$

$$28 = \frac{40}{360}\pi R^2 - \frac{1}{2}R^2 \sin 40^\circ$$

$$28 = R^2\left(\frac{\pi}{9} - \frac{1}{2}\sin 40^\circ\right)$$

$$R = 31.8096\ldots$$

4. Use trigonometry

Now find the chord BC:

$$BC = 2R\sin 20^\circ$$

$$BC = 2(31.8096\ldots)\sin 20^\circ = 21.7591\ldots$$

5. Calculate value

In sector BCD, the centre is C, so the radius is

$$CB = CD = 21.7591\ldots$$

6. Use trigonometry

The arc BD has angle 130°, so

$$\text{arc } BD = \frac{130}{360} \times 2\pi(21.7591\ldots)$$

$$= 49.3697\ldots$$

FINAL ANSWER

$\boxed{49.4 \text{ cm}}$

PAST PAPER QUESTION

Q#: 25 Marks: 6

TOPIC: **Circles, Arcs & Sectors**

Jun 2025 P1HR - Jun 2025 | P1HR

25

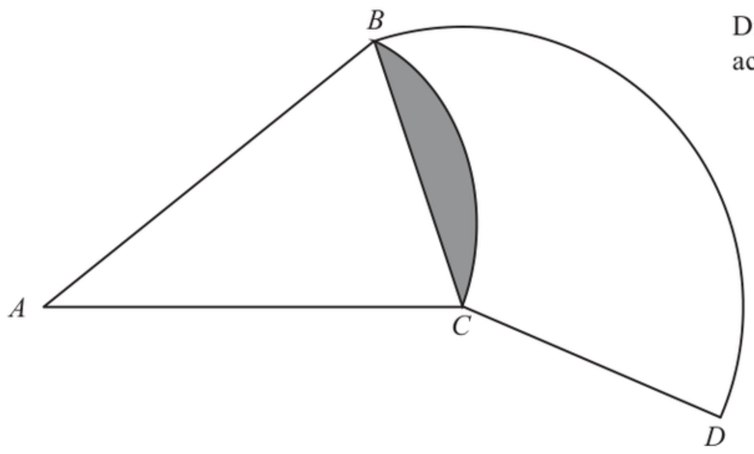

Diagram **NOT** accurately drawn

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Area of shaded segment $= 28\,\text{cm}^2$

Find the length of the arc BD
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 25 is 6 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 6

TOPIC: **Circles, Arcs & Sectors**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Let the radius of sector

Let the radius of sector BAC be R.

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The shaded segment is the segment cut off by chord BC in the sector with centre A.

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$$\text{sector area} - \text{triangle area}$$

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$$= 49.3697\ldots$$

FINAL ANSWER

$\boxed{49.4 \text{ cm}}$

PAST PAPER QUESTION

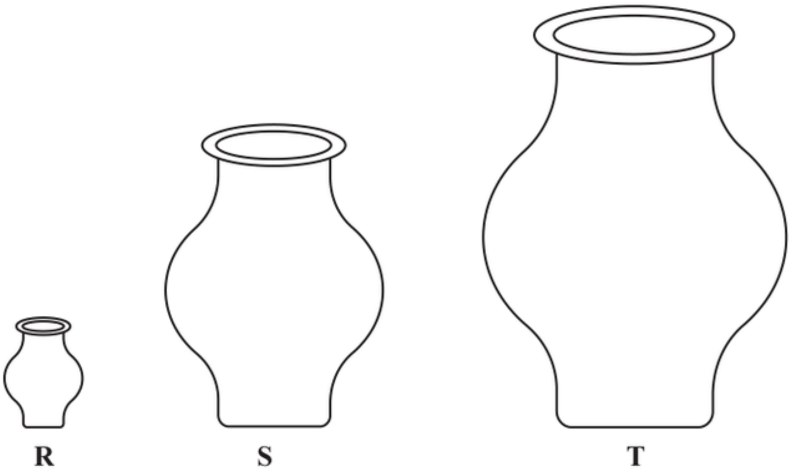
Q#: 26 **Marks: 4**

TOPIC: **Area & Volume of Similar Shapes**

Jun 2025 P1HR - Jun 2025 | P1HR

26 **R**, **S** and **T** are three similar vases.

Diagram **NOT** accurately drawn



The volume of vase **S** is 72.8% more than the volume of vase **R**

The height of vase **R** is h cm
The height of vase **T** is $6h$ cm

The surface area of vase **S** is A cm^2
The surface area of vase **T** is kA cm^2

Work out the value of k

k = ...

(Total for Question 26 is 4 marks)

PAST PAPER SOLUTION

Q#: 26 Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. The volume of vase is

The volume of vase S is 72.8% more than vase R, so

$$\frac{V_S}{V_R} = 1.728$$

2. Calculate area

For similar solids, volume scale factor is the cube of the linear scale factor:
linear scale factor from R to $S = \sqrt[3]{1.728} = 1.2$

3. The height of vase is

The height of vase T is $6h$, so the linear scale factor from R to T is 6.

4. Calculate area

Therefore the linear scale factor from S to T is

$$\frac{6}{1.2} = 5$$

5. Surface area scale factor is

Surface area scale factor is the square of the linear scale factor:
$k = 5^2 = 25$

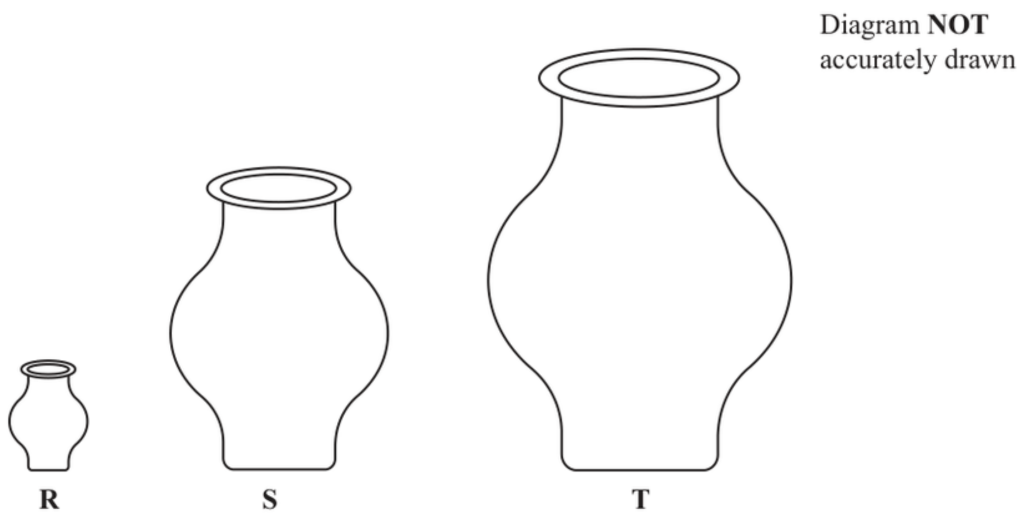
FINAL ANSWER
$k = 25$

PAST PAPER QUESTION Q#: 26 Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Jun 2025 P1HR - Jun 2025 | P1HR

26 **R**, **S** and **T** are three similar vases.



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Work out the value of k

k = ...

(Total for Question 26 is 4 marks)

PAST PAPER SOLUTION

Q#: 26 Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

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FINAL ANSWER

$k = 25$